Ant War: Game Design Document

*Game jam draft v1 (2-day scope). Items marked **[PROPOSED]** were not in the original notes and need your approval.*

1. Concept

Ant War is a real-time 2D strategy game with a Mount & Blade style world map. The player picks an ant species (class), then expands across a map of anthills by scouting, attacking and holding territory. Ants never die. They get injured and need time to recover, so every attack leaves your own colony exposed.

Genre: Real-time map strategy **Perspective:** 2D top-down world map **Session length [PROPOSED]:** 15 to 25 minutes per run

2. Design Pillars

1. **Risk vs. reward.** Sending ants out is always a gamble because the home hill is left weak.
2. **Scout before you strike.** Information is the main resource. You see targets before you attack them.
3. **Relationships matter.** Attacking one species can bring its allies down on you.
4. **Nobody dies.** Losses are measured in downtime, not casualties.

3. Game Flow

1. Title screen
2. Class (species) select
3. World map (all gameplay happens here)
4. Win or lose screen, then restart or back to the title screen

4. Class Selection

The player chooses a species, like choosing a class in a soulslike. Each species has different starting stats and diplomacy.

Example: Bullet ants have strong attack and low numbers.

[PROPOSED] Stats per species (keep to 4 for the jam):

Stat	Effect
Attack	Power per ant in battle
Defense	Power per ant when defending
Recovery speed	How fast injured ants return to action
Growth rate	How fast the hill's max colony size grows over time

[PROPOSED] Species count: 4 playable and 4 total on the map for the jam. For example Bullet Ants, Fire Ants, Army Ants and Weaver Ants. Add more only if time allows.

5. World Map

- A single 2D map with fog of war: anthills are hidden until scouted.
- The player's starting anthill is visible from the start.
- Anthills are placed at fixed positions, with a number of ants shown above each hill.
- **[PROPOSED]** Map layout is hand-placed (no procedural generation), about 8 to 12 anthills total.
- **[PROPOSED]** The map is a single screen or a small scrolling area with a camera that follows the scout.

6. Anthill Menu

Clicking an anthill opens a menu. This pauses the game.

Own anthill:

- Number of ants in the colony (available, away, recovering)
- **Diplomacy panel:** which species are allied and which are hostile, to the player and to each other
- **Send out scout** button
- **Recruit scout** button (only when a scout is present, see section 7)

Scouted enemy or ally anthill:

- Species living there
- Ant count
- Their allies
- **Attack** button

Anthill Buffs

Every anthill has one buff that applies to the colony living there. Whoever owns the hill gets the buff, so capturing a hill is also a way to pick up a perk.

Buff	Effect
Healing Grounds	Injured ants recover faster
Fortified	Defending ants get a defense bonus
Warren	Colony grows faster
War Camp	Ants sent from here get an attack bonus
Lookout	Scouts from here move faster or reveal more of the map

[PROPOSED] Buffs are hand-assigned per hill (no randomness), shown in the anthill menu, and carry over to the new owner on capture. Start with 3 or 4 buffs for the jam and add the rest only if time allows.

Colony Growth

Every anthill slowly grows its max ant capacity over time, which grows its colony. Each species has its own growth rate, so some species snowball faster than others.

- Growth is slow and continuous, running in real time.
- Growth rate comes from the hill's species stat and can be boosted by the Warren buff.
- **[PROPOSED]** Ant count fills up toward the current max capacity at a steady rate, so injured ants and sent-out ants still count toward the colony.
- **[PROPOSED]** Ants gained from a capture that exceed the new hill's capacity stay in the colony but don't add extra growth.
- This applies to NPC hills too, so waiting too long makes the map harder. It acts as a soft timer on the player.

7. Scouts

A scout is a controllable unit on the world map.

- Sent from an anthill via its menu.
- Click on the map to move it. It reveals anthills it gets near.
- Clicking a scouted anthill opens its info menu.

- **Multi-hill attacks:** if the player owns several anthills, they take the scout to another owned hill and click **Recruit scout**. A new scout spawns and follows the first. Repeat for as many hills as needed.
- When the player clicks **Attack**, every scout returns to its own home hill. As each arrives, the player is prompted for how many ants to send from that hill.
- Moving a single scout straight to a target means only its home hill is asked.

[PROPOSED] Scout rules:

- Scouts are invincible while roaming the map. Nothing can catch, injure or block them, and they only fight when an army attacks an anthill or defenders hold one.
- Only one scout trip per hill at a time.
- NPC scouts are visible to the player so they can react.

8. Attack Flow

1. Scout reaches the target and the player opens its menu.
2. Player clicks **Attack**.
3. Scouts return home.
4. On arrival at each home hill, a slider prompts "How many ants to send?" (this pauses the game).
5. Ants march to the target (travel time).
6. Battle resolves (see section 9).
7. Ants return home and enter recovery.

The core tradeoff: more ants means a better chance to win but a weaker home defense. Fewer ants means a safer home but a bigger share of the sent ants get injured.

Because marching, fighting and returning all take time, the home hill is exposed for a long window.

9. Battle Resolution (not in the original notes)

This needs a decision, because it is the heart of the game. **[PROPOSED]** for jam scope: automatic and instant resolution when the army arrives, shown as a short animation or timer bar.

Formula:

- Attacker power = sent ants x Attack stat
- Defender power = healthy ants x Defense stat (+ a defender bonus, for example 10 to 20 percent)

- Higher power wins. Add a small random factor (about plus or minus 10 percent) so it isn't fully deterministic.
- Injury count for each side scales with the opponent's power, for example the loser has most of its ants injured and the winner a smaller share.

The winner of an attack captures the hill. Defenders at that hill join the attacker's colony (see section 11).

10. Injury and Recovery

- Ants never die.
- Injured ants are inactive: they can't attack or defend.
- Each injured ant has a recovery timer. **[PROPOSED]** The timer is per batch, not per ant, to keep the data simple.
- Injuries apply to both winners and losers, and to defenders when a hill is attacked.
- Recovery speed is a species stat.

Example: a hill with 10 ants is attacked and 6 are injured, so only 4 can defend until the injured ants recover.

11. Capturing Anthills

- A hill is capturable when the player can send more power than its defenders.
- On capture, the defending ants join the player's colony.
- **[PROPOSED]** Captured ants arrive as injured/recovering, so a fresh capture isn't instantly strong.
- **[PROPOSED]** The hill's species becomes the owner's, but ants keep their original species stats. This is optional for the jam. Simplest: all ants use the owner's class stats.
- Captured hills can spawn scouts and send attacks like the home hill, and they bring their buff and growth with them.

12. Diplomacy and Retaliation

- Every species has a relationship with every other species: **ally**, **hostile** or **neutral**.
- Relations are fixed at the start of the game (no diplomacy actions for the jam).
- The diplomacy info panel in the anthill menu shows these relations.

Ally retaliation: when species A is attacked, A's allies see the player's colony is weakened (ants are out attacking) and may launch their own attack on the player's hill to retaliate.

[PROPOSED] Rule: when an attack on A starts, each ally of A rolls a chance to send a retaliation scout toward the nearest player hill. A retaliation attack happens only if the player's home defense looks weaker than the ally's available force.

This is the reason players plan around relations before attacking.

13. NPC Behavior

NPC anthills follow the same rules as the player:

1. Send a scout toward a target (visible to the player).
2. Scout reaches the target and checks the numbers.
3. If attack is favorable, scout returns, NPC picks how many ants to send.
4. Army marches, fights, returns and recovers.

[PROPOSED] Simple AI for jam scope:

- Each NPC hill has a decision timer (for example every 30 to 60 seconds).
- It picks the weakest visible neighbor (preferring hostile species).
- It sends if its available power exceeds the target's by a margin, and keeps a minimum reserve at home.
- Retaliation (section 12) is a separate trigger.

NPC difficulty can be tuned with the decision timer and the margin.

14. Win and Lose Conditions

- **Win:** the player owns every anthill on the map.
- **Lose:** every anthill the player owns has been captured.

[PROPOSED] Show a short summary on end: time played, hills captured, hills lost.

15. Real-Time and Pause

- The game runs in real time.
- Only these pause the game: anthill menus, the send-ants prompt and the pause menu.
- **[PROPOSED]** Add a speed control (1x, 2x) to make waiting less tedious. This is optional and low effort.

16. UI Checklist

- Class select screen

- World map HUD: total ants, number of hills owned, pause button
- Ant count labels above every anthill (scouted hills only)
- Anthill menu (own and scouted variants)
- Diplomacy panel
- Send-ants slider prompt
- Battle result popup (win or loss, injuries)
- Pause menu
- Win and lose screens
- **[PROPOSED]** Alert for "Your hill is under attack" and "Enemy scout spotted" (essential, since everything is real-time)

17. Art and Audio Asset List (for your teammate)

Art:

- Species icons or portraits (class select), 4 total
- Anthill sprite (one base, tinted per species)
- Scout ant sprite (walk cycle optional)
- Army marker sprite (a group of ants moving on the map)
- World map background
- UI frames, buttons and sliders
- Injured and recovering status icon
- Fog of war overlay

Audio (optional but cheap): click, attack start, battle result sting, alert ping, one looping music track.

18. Jam Scope

Must have (MVP):

- Class select with 3 to 4 species
- World map with fog of war and a hand-placed anthill set
- Scout movement and anthill info menu
- Attack flow with send-ants prompt and automatic battle resolution
- Injury and recovery system
- Win and lose conditions
- Basic NPC AI (attack the weakest neighbor)

Should have:

- Multi-scout recruitment from several hills
- Ally retaliation
- Diplomacy panel
- Alerts for incoming attacks

Nice to have:

- Speed control
- Species-specific abilities
- Sound and music polish
- Battle animations

Cut if short on time: mixed-species ants after capture, procedural maps, extra species.

19. Suggested 2-Day Schedule

Day 1:

- Core data model: anthill (owner, species, ants, injured batches), species stats
- World map, scout movement, fog of war, anthill menu
- Attack flow end to end against one dumb target

Day 2 morning:

- Battle resolution, injury and recovery, win and lose
- NPC AI with the same scout-and-attack rules

Day 2 afternoon:

- Ally retaliation, multi-scout recruitment
- Alerts, balance pass, art integration, bug fixing

Last hour: build, playtest, ship. No new features.

20. Open Questions and Risks

1. **What happens when a scout is caught or stranded?** Resolved: scouts are invincible while roaming and only fight during attacks or defenses.
2. **Can an NPC attack a hill while it is already under attack?** Suggest yes, which is how chain captures happen.
3. **Can the player cancel an attack while ants are marching?** Suggest no for the jam.

4. **What happens to ants if their home hill is captured while they are away?** Suggest they return to the nearest owned hill, or are absorbed by the capturer if none.
5. **Is there a cap on how fast ants regenerate or reproduce?** Without growth, the game is a pure zero-sum shuffle. Resolved: every hill slowly grows its max capacity, at a rate set by species (see Colony Growth).
6. **Risk: real-time with long waits.** Marches and recoveries can feel slow. Tune speeds early and add a speed control.
7. **Risk: scope.** The multi-scout and ally retaliation systems are the most complex parts. Build the single-scout version first, and only then add them.